Physics-Informed Federated Safe Reinforcement Learning for Bidirectional V2G Coordination

A Differentiable Residual-Dyna Architecture with Hard Safety Layers and Willingness-Aware Federated Bidding

Abstract

This research specification proposes a Differentiable Residual-Dyna World Model ($\hat{s}_{t+1} = F_{base}(s_t, a_t) + N_\theta(s_t, a_t)$) for Vehicle-to-Grid (V2G) power distribution network control. By replacing black-box neural surrogates (e.g. static PINNs) with an explicit analytical AC power-flow base model (F_{base}) coupled to a neural residual network (N_θ), the framework is designed to enable unbroken end-to-end analytical gradient backpropagation ($\partial F_{base}/\partial a_t$) into the reinforcement learning policy. Combined with a Quadratic Programming (QP) hard-safety projection layer and dynamic User V2G Participation Willingness (VPW) bidding, the proposed system targets zero-violation operation (no voltage or thermal breaches), an estimated on-the-order-of 50% reduction in charging cost, and an expected 4x–10x acceleration in sample convergence relative to the PINN and soft-penalty Safe RL baselines, all to be established experimentally rather than assumed. Feasibility of the full architecture at 123-bus scale is contingent on a preliminary QP-runtime benchmark (Section 4.1); the 33-bus feeder is the primary, low-risk deliverable, with 123-bus treated as a stretch goal.

Keywords: Vehicle-to-Grid (V2G), Differentiable Physics, Residual Dyna Architecture, Safe Reinforcement Learning, Quadratic Programming Projection, Personalized Federated Learning.

1 Re-Engineered Problem Statement: The Triple Dilemma

The rapid adoption of bidirectional V2G technology transforms electric vehicle (EV) fleets into distributed battery energy storage systems capable of providing peak-shaving, frequency regulation, and localized voltage support to active distribution networks (ADNs). Scaling V2G coordination across many dynamic EV nodes exposes a Triple Dilemma of Scale, Safety, and Privacy that state-of-the-art RL and mathematical-optimization approaches do not resolve simultaneously:

- **The Computational & Gradient Bottleneck:** Exact AC Optimal Power Flow (AC-OPF) is non-convex and NP-hard to compute at sub-second operational intervals. Existing data-driven surrogate models (e.g. offline PINNs) act as uncoupled black boxes that block the backpropagation gradient chain (∂Physics/∂Action), causing compounding prediction error over multi-step model-based rollouts.
- **The Exploration–Safety Dilemma:** Conventional Safe RL relies on soft-penalty reward shaping. During exploration, or under unexpected solar/demand fluctuations, these agents can still commit physical violations – feeder voltage sag (< 0.95 p.u.), voltage swell (> 1.05 p.u.), or transformer thermal overload.

- **The Heterogeneous Non-IID / Privacy Barrier:** Charging stations operate in heterogeneous environments (residential overnight vs. commercial workplace) with diverse user travel needs, battery-degradation friction, and V2G Participation Willingness (VPW). Standard FedAvg diverges under severe Non-IID client distributions. Centralized data collection also raises privacy concerns; Section 5.4 scopes precisely what this proposal does and does not claim to defend against.

2 Mathematical Foundations & Physical Grid Equations

The grid is modeled as a radial distribution network $G = (V, E)$, $V = \{0, 1, \ldots, N\}$ (bus 0 is the substation slack bus), E the set of feeder branches.

2.1 Full AC Power Flow & LinDistFlow Branch Equations

$$P_i(t) = V_i(t) \sum_{j \in N_i} V_j(t)\big(G_{ij} \cos\theta_{ij}(t) + B_{ij} \sin\theta_{ij}(t)\big) \tag{1}$$

$$Q_i(t) = V_i(t) \sum_{j \in N_i} V_j(t)\big(G_{ij} \sin\theta_{ij}(t) - B_{ij} \cos\theta_{ij}(t)\big) \tag{2}$$

For radial feeders, the exact DistFlow recursion (active power P_{ij}, reactive power Q_{ij}, squared current I_{ij}^2, squared voltage $v_i = V_i^2$) along line (i, j):

$$P_{ij,t+1} = P_{jk,t+1} + r_{ij} I_{ij,t+1}^2 + P_{L,j,t+1} - P_{V2G,j,t+1} \tag{3}$$

$$Q_{ij,t+1} = Q_{jk,t+1} + x_{ij} I_{ij,t+1}^2 + Q_{L,j,t+1} - Q_{V2G,j,t+1} \tag{4}$$

$$v_{j,t+1} = v_{i,t+1} - 2(r_{ij} P_{ij,t+1} + x_{ij} Q_{ij,t+1}) + (r_{ij}^2 + x_{ij}^2) I_{ij,t+1}^2 \tag{5}$$

$$I_{ij,t+1}^2 = \frac{P_{ij,t+1}^2 + Q_{ij,t+1}^2}{v_{i,t+1}} \tag{6}$$

2.2 Linearized Voltage Sensitivity Matrix (LinDistFlow)

Under the first-order LinDistFlow linearization ($v_i \approx 1.0$ p.u., $(r_{ij}^2 + x_{ij}^2) I_{ij}^2 \approx 0$), we obtain an explicit linear mapping from net power injections to bus voltage magnitudes:

$$v(t) = v_0 + R\, P_{net}(t) + X\, Q_{net}(t) \tag{7}$$

where $v \in \mathbb{R}^N$, $P_{net} = P_{V2G} - P_L$, $Q_{net} = Q_{V2G} - Q_L$, and $R, X \in \mathbb{R}^{N \times N}$ are the constant voltage sensitivity matrices of the *linearized* model, derived directly from the bus admittance matrix Y_{bus}. R and X are exact within this linearized approximation; they are not exact relative to the full non-linear AC formulation in Eqs. (1)–(2), and the residual network N_θ (Section 3) is precisely what is tasked with recovering the gap between the two.

2.3 Battery State-of-Charge (SoC) & Degradation

$$SoC_k(t+1) = SoC_k(t) + \left(\eta_{ch} P_{k,t}^{ch} - \frac{1}{\eta_{dis}} P_{k,t}^{dis}\right) \frac{\Delta t}{E_k^{cap}} \tag{8}$$

$$C_{deg}(P^{dis}) = \beta \left(P_{k,t}^{dis}\right)^{\gamma} + \gamma \left(1 - SoC_k(t)\right)^2 \tag{9}$$

3 The Differentiable Residual-Dyna World Model Architecture

$$\hat{s}_{t+1} = F_{base}(s_t, a_t) + N_\theta(s_t, a_t) \tag{10}$$

F_{base} is the explicit analytical physics engine (LinDistFlow + SoC dynamics); N_θ is a neural network learning the unmodeled residual $\Delta s = s_{true} - F_{base}$.

3.1 Exact Analytical Jacobian of the Base Model

$$\frac{\partial F_{base}}{\partial a_t} = \begin{bmatrix} R & X \\ \dfrac{\eta_{ch}\Delta t}{E^{cap}} & -\dfrac{\Delta t}{\eta_{dis}E^{cap}} \end{bmatrix} \tag{11}$$

This Jacobian is exact for F_{base} (as defined under the LinDistFlow linearization) and computationally instant, eliminating finite-difference approximation.

3.2 Per-EV to Bus-Level Aggregation

Each bus i may host multiple simultaneously-connected EVs, indexed by the set $K_i(t)$. The bus-level net injections used in Eq. (7) are the sum of individual EV actions at that bus, net of local load:

$$P_{net,i}(t) = \sum_{k \in K_i(t)} P_k^{V2G}(t) - P_{L,i}(t) \tag{12}$$

$$Q_{net,i}(t) = \sum_{k \in K_i(t)} Q_k^{V2G}(t) - Q_{L,i}(t) \tag{13}$$

Three properties of this aggregation step matter for the rest of the design:

1. **Contention.** When $|K_i(t)| > 1$ and the bus-level safety projection (Section 4) tightens the feasible region, the QP projection redistributes the correction across the EVs at that bus; VPW (Section 5) determines the priority order for who absorbs the correction.

2. **Privacy.** The sum $P_{net,i}$ is computed *locally at the charging station* before any federated communication round – individual EV session data (P_k^{V2G}, SoC_k) never leaves the station; only the aggregated bus-level quantity and the (locally updated) FedProx head weights are exposed to the coordination layer.

3. **Differentiability.** Because the aggregation is a linear sum, $\partial P_{net,i}/\partial P_k^{V2G} = 1$ for all $k \in K_i(t)$, so the Jacobian in Eq. (11) extends cleanly to the multi-EV setting without approximation.

3.3 Unbroken Policy Gradient Backpropagation

$$\nabla_\phi J(\phi) = \mathbb{E}_{s_t}\left[\nabla_a Q_\psi(s_t, a)\big|_{a=\pi_\phi(s_t)} \cdot \left(\frac{\partial F_{base}}{\partial a_t} + \frac{\partial N_\theta}{\partial a_t}\right) \cdot \nabla_\phi \pi_\phi(s_t)\right] \tag{14}$$

$$\mathcal{L}_{World}(\theta) = \frac{1}{|B|} \sum_{(s_t, a_t, s_{t+1}) \in B} \left\| s_{t+1} - \left(F_{base}(s_t, a_t) + N_\theta(s_t, a_t)\right)\right\|_2^2 + \lambda_{phys}\mathcal{L}_{KCL/KVL}(N_\theta) \tag{15}$$

4 Hard Constraint Safety Layer & Action Masking

$$a_t^{safe} = \arg\min_{a \in A} \frac{1}{2} \|a - a_t^{raw}\|_2^2 \tag{16}$$

$$\text{s.t.} \quad V_{min} \leq v_0 + RP_{net}(s_t, a) + XQ_{net}(s_t, a) \leq V_{max} \tag{17}$$

$$I_{ij}(s_t, a) \leq I_{ij,max}, \quad \forall (i,j) \in E \tag{18}$$

Solved via implicit differentiation (OptNet / CVXPYLayers), keeping the projection differentiable so the Actor learns the boundary of the physically safe action space.

4.1 Computational Scalability & QP Runtime Benchmarking Plan

The QP in Eqs. (14)–(16) must be solved once per control timestep, per feeder, via a forward CVXPYLayers pass at inference time and a forward+backward pass during training. Solve cost scales with the number of buses N (through $R, X \in \mathbb{R}^{N \times N}$ and the $2N$ voltage-bound constraints) and the number of branches $|E|$ (thermal constraints), so 33-bus and 123-bus are not expected to be equally cheap.

Plan. Before committing the full team structure to 123-bus scope, run a short, isolated benchmark (target: Week 2 of the semester) that measures, independently of the rest of the pipeline:

Feeder	# constraints	Forward solve time	Forward+backward time
IEEE 33-bus	$2N + \|E\|$	X_{33} ms (TBD — Week 2 benchmark)	X'_{33} ms (TBD)
IEEE 123-bus	$2N + \|E\|$	X_{123} ms (TBD — Week 2 benchmark)	X'_{123} ms (TBD)

Table 1: Preliminary QP runtime benchmark template. Cells are left as placeholders to be filled in after running the benchmark described below.

Method. Instantiate the R, X admittance-derived matrices for the IEEE 33-bus and 123-bus test feeders (available directly from pandapower's built-in network library, no need to wait on the rest of the physics engine), formulate the QP of Eqs. (14)–(16) in CVXPYLayers with a representative number of concurrently-connected EVs drawn from ACN-JPL session-overlap statistics (e.g. batches of 50, 200, and 500 concurrent sessions), and measure wall-clock forward and forward+backward solve time on the target training hardware, warm-started and cold-started.

Decision rule. Compare the measured 123-bus forward+backward time against the intended control-loop period (e.g. a 1–5 minute V2G dispatch interval leaves generous headroom; a sub-second real-time loop would not). If 123-bus solve time is within budget, proceed with 123-bus as a stretch target; if not, batch the QP over EV subsets or fall back to 123-bus as an evaluation-only, reduced-frequency scenario. In either case, **33-bus is the primary, committed deliverable**, consistent with the low-risk core of the project (Feasibility Verdict).

Sequencing. This benchmark does not block Student 1's physics-engine work (Section 7.2), which starts in parallel; it blocks only the decision to lock 123-bus into the committed scope. Once the benchmark and the choice of primary/stretch scale are settled (target: end of Week 2), the team proceeds directly into full implementation rather than revisiting methodology mid-semester.

5 Driver Willingness (VPW) & Personalized Federated Learning

5.1 Dynamic V2G Participation Willingness (VPW) Bidding Model

$$W_k(t) = w_1 \cdot \frac{T_k^{park}(t)}{T_k^{req}} + w_2 \cdot \frac{SoC_k(t) - SoC_k^{min}}{SoC_k^{target} - SoC_k^{min}} + w_3 \cdot \frac{\lambda_{grid}(t) - \lambda_{avg}}{\lambda_{max} - \lambda_{min}} - w_4 \cdot \frac{C_{deg}(P_{k,t}^{dis})}{Rev_{max}} \quad (19)$$

5.2 Weight Calibration Procedure

Since ACN-JPL contains no direct willingness label, $w_1 \dots w_4$ are calibrated in two stages:

1. **Fixed-heuristic baseline.** Initialize weights via grid search against a proxy signal derived from revealed behavior in the dataset – historical session opt-in rate and early-departure (opt-out) frequency conditioned on parking duration, SoC, and time-of-use price, used as a stand-in for latent willingness.
2. **Stretch goal.** If time permits, replace the fixed heuristic with per-user learned weights fit via inverse RL against the same proxy signal, evaluated for correlation with the fixed-heuristic ranking as a sanity check rather than assumed superior by default.

This is treated as a secondary, non-load-bearing contribution: results should not hinge on VPW outperforming a naive FCFS or SoC-only prioritization baseline, since the proxy signal is not a validated ground truth.

5.3 Personalized Federated Learning with Local Decoupling (FedProx)

$$\min_{w_i} \mathcal{L}_i(w_i) + \frac{\mu}{2}\|w_i - w_{global}^t\|_2^2 \quad (20)$$

$$w_i = [w_i^{shared}, w_i^{head}], \quad w_{global}^{t+1} = \sum_{i=1}^{M} \frac{N_i}{N_{total}} w_i^{shared,t+1} \quad (21)$$

Shared feature extractor layers w^{shared} aggregate centrally via FedAvg; local decision heads w^{head} remain at station i for personalization.

5.4 Privacy Scope

FedProx with parameter decoupling addresses *Non-IID convergence*, not gradient-inversion resistance. This proposal's committed claim is scoped to Non-IID robustness: personalized FedProx converges under heterogeneous client distributions where FedAvg diverges. *Gradient-inversion resistance is not claimed by the core design.* As an optional hardening layer, if time allows, DP-SGD noise injection on the shared-gradient upload (or secure aggregation) can be added and evaluated as a stretch goal; Table 2 reflects this narrower, honest claim rather than an unsubstantiated "Privacy" column.

6 Comparative Research Literature Benchmarking Matrix

7 Experimental Setup, Datasets, Ablations & Timeline

7.1 Benchmark Datasets & Software Stack

- EV Session Logs: Caltech/JPL ACN Dataset.

Paper/Study	Physics Grounding	Gradient Flow	Safety
Gong et al. (2026) [cite]	Building load limits + Fed. PPO	Blocked (Discrete)	Soft Penalty
Qian et al. (FedSAC) [cite]	External OPF solver + FedSAC	Blocked (External)	Hard OPF
PI-TD3 [cite: pinn.pdf]	Diff. Power Flow + TD3	Unbroken (Exact)	Soft Penalty
Dynamic Reconfig. [cite]	Spatiotemporal GCN + IPPO	Blocked (Implicit)	Action Maski
Hybrid 3 (Mansour) [cite]	XGBoost + BiLSTM	N/A (Forecast)	None
EV2Gym (2024) [cite]	OpenDSS/pandapower Gym Benchmark	External Simulator	Custom
PINN.docx (Original) [cite]	Static PINN Surrogate + Safe RL	Blocked (Black-box)	Soft Penalty
Proposed Framework	Differentiable Residual Dyna	Unbroken (Exact $\partial F/\partial a$)	Hard QP (target: 0 v

Table 2: Comparative benchmarking matrix. See Table 3 for trade-off considerations.

Dimension	Proposed Framework	Typical Baselines
Per-step compute overhead	Higher (online QP solve + backward pass)	Lower (no online QI
Dependence on accurate Y_{bus}	Yes (F_{base} requires it)	No (purely data-driven ba
Requires topology re-derivation on feeder change	Yes (R, X recomputed)	Varies

Table 3: Trade-off considerations relative to baseline approaches.

- Power Distribution Test Feeders: IEEE 33-bus and IEEE 123-bus, in OpenDSS and pandapower.
- Household Demand & Solar: Pecan Street smart meter data; NREL NSRDB, 15-min resolution.
- Simulation: EV2Gym + PyTorch 2.x, CVXPYLayers, Flower/Ray.

7.2 Ablation Plan

1. F_{base}-only vs. $F_{base} + N_\theta$ – isolates the residual network's contribution.
2. Soft-penalty Safe RL vs. QP hard-safety projection – isolates the safety layer's contribution.
3. FedAvg vs. personalized FedProx – isolates the federated-learning contribution.
4. With vs. without VPW prioritization – isolates VPW's contribution, evaluated against the FCFS/SoC-only baseline noted in Section 5.2.

7.3 Team Work Distribution (3 Researchers)

- **Student 1 (Power Systems & Base Physics Engine):** DistFlow/LinDistFlow formulation, OpenDSS/pandapower digital twins for IEEE 33/123-bus, extraction of R, X, the analytical F_{base} engine.
- **Student 2 (RL & Hard Safety Layers):** N_θ implementation, TD3/PPO Actor-Critic, the CVXPYLayers QP safety layer (including the scalability benchmark in Section 4.1), the Dyna planning rollout loop.
- **Student 3 (Federated Learning, Willingness & Benchmarking):** VPW bidding and its calibration procedure (Section 5.2), personalized FedProx pipeline, cross-site benchmarking on ACN/Pecan Street, and the ablation runs in Section 7.2.

7.4 Semester Timeline

- **Weeks 1–2:** QP scalability benchmark (Section 4.1); lock 33-bus vs. 123-bus scope; begin F_{base} implementation (Student 1) in parallel.
- **Weeks 3–6:** $F_{base} + N_\theta$ world model on 33-bus; TD3/PPO loop skeleton (Students 1–2).
- **Weeks 5–8:** QP safety layer integration and differentiable projection (Student 2).
- **Weeks 6–10:** FedProx pipeline, VPW proxy calibration (Student 3).
- **Weeks 9–12:** Full 33-bus integration; ablation runs.
- **Weeks 11–14:** 123-bus attempt, conditional on Week 2 benchmark result.
- **Weeks 14–16:** Final benchmarking table, write-up, submission.

8 Conclusion

This revision establishes a scoped, defensible blueprint for scalable, safe, and Non-IID-robust bidirectional V2G coordination. The core contribution – differentiable, physically-grounded gradient flow through a Residual-Dyna model with hard QP safety – is a combination of established, working techniques rather than new theory, which is the safer profile for a B.Tech-scope project. Its main open risk, 123-bus QP scalability, is resolved by measurement rather than assumption (Section 4.1) before the team commits to that scope.

References

Citation details below are placeholders (`TBD`) for the six papers named in Table 2; replace with full bibliographic details before submission.

References

[1] Gong et al. (2026) – full citation TBD.

[2] Qian et al., FedSAC. Full citation TBD.

[3] PI-TD3. Full citation TBD (source: `pinn.pdf`).

[4] Dynamic Reconfiguration (Spatiotemporal GCN + IPPO). Full citation TBD.

[5] Hybrid 3 (Mansour). Full citation TBD.

[6] EV2Gym (2024). Full citation TBD.